Handwriting Recognition System

Project Overview

The **Handwriting Recognition System** is a deep learning-based application designed to detect and compare handwritten text. The system uses convolutional neural networks (CNNs) to extract features from handwriting samples and determine similarities between documents.

This project is particularly useful in **plagiarism detection, document verification, and handwriting analysis**.

Problem Statement

Manual verification of handwritten documents can be difficult and time-consuming. Educational institutions and organizations often require automated systems to identify similarities between handwritten texts.

Objectives

- Build an automated system to analyze handwriting patterns.
- Compare two handwritten documents.
- Detect similarity between handwriting samples.

Technology Stack

Programming Language-Python

Deep Learning Framework- TensorFlow , Keras

Computer Vision- OpenCV

Machine Learning- Convolutional Neural Networks (CNN), Feature Extraction

Data Processing- NumPy, Pandas

Model Evaluation- Scikit-learn

Visualization- Matplotlib, Seaborn

Tools-Jupyter Notebook

Dataset / Input

- Images of handwritten text
- Scanned handwritten documents

Methodology / Working

1. Handwritten images are collected.

2. Images are preprocessed using OpenCV:
    - Noise removal
    - Grayscale conversion
    - Normalization
3. CNN extracts handwriting features.
4. Feature vectors are compared.
5. The system calculates similarity scores.

Key Features

- Handwriting pattern recognition
- Feature extraction using CNN
- Image preprocessing using OpenCV
- Document similarity detection

Results / Output

The model successfully detects similarities between handwritten documents, helping identify possible plagiarism or matching handwriting patterns